Fibonacci Sum

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Problem 0.1 – Fibonacci Sum Write F_n for the n-th Fibonacci Number, with $F_1 = F_2 = 1$ and $F_n = F_n + F_{n-1}$. It is well known that F_n is very well approximated by $\frac{\phi^n}{\sqrt{5}}$, where ϕ, the golden ratio, is the positive root of the equation $x^2 = x + 1$.

Let $G(n)$ be the number of distinct integers $0 \leq x \leq n$ such that $x^2 \equiv x + 1 \pmod{n}$.
You are given:

$$\sum_{n=1}^{10^3} F_n G(n) = 190950976 \pmod{10^9 + 9}$$

Find

$$\sum_{n=1}^{10^{14}} F_n G(n)$$

giving your answer modulo $10^9 + 9$.

0.1 Establishing the Congruence

The first thing to do is reestablish $G(n)$ as the significantly easier to solve $x^2 = n \pmod{z}$. This can be done by completing the square:

$$\begin{aligned} x^2 - x - 1 &= 0 \\ x^2 - x &= 1 \\ 4x^2 - 4x + 1 &= 5 \\ (2x - 1)^2 &= 5 \end{aligned}$$

Functionally, counting the solutions to $G(n)$ is equivalent to counting the square roots of 5 modulo n. This is doubly assured by the problem's note of Binet's formula above, which requires $\sqrt{5} \pmod{5}$ to exist.